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Axial Symmetry Directs Ligand Orientation in Galectin-3

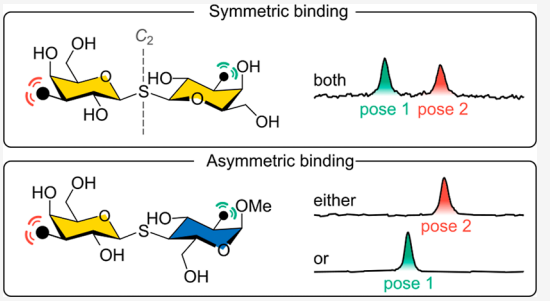
Nina Habanová,*,‖ Jakub Zýka,‖ Vít Prouza, Jaroslav Kozák, Martin Zavřel, Milan Kožíšek, Pavel Srb, Jan Choutka,*,‖ Kamil Parkan,* and Radek Pohl

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ABSTRACT: Programmatic control over bound ligand poses remains a significant challenge in drug design due to the lack of universal design principles that could be harnessed to direct the ligand orientation in the binding site. Here, we demonstrate that molecular symmetry, combined with the high specificity of lectin−carbohydrate recognition, can be utilized to control the binding of small molecules to galectin-3. We synthesized a series of galectin-3 ligands that differ in the C_2-rotational symmetry features of their saccharide cores, imposing direct control over their binding behavior. Using a combination of ligand-observed and protein-observed NMR spectroscopy, complemented by free energy calculations, we were able to comprehensively describe the symmetry-resolved binding poses within this ligand series as well as the underlying thermodynamics. Our work offers new strategies for designing galectin-3 inhibitors with predictable binding behavior and the delivery of substituents to defined locations within the binding pocket.



KEYWORDS: *Galectin-3, ligand binding poses, molecular symmetry, glycomimetics, NMR spectroscopy, free energy perturbation*

INTRODUCTION

Ligands can bind to their macromolecular receptors in one or more orientations (aka binding poses). The ability to predict, describe, and control these poses remains an outstanding task in structural biology and drug design[1]—one made especially important given that different ligand poses within a receptor pocket often affect the affinity, selectivity, and even functional outcome (e.g., agonism vs antagonism) of the interaction.[2−6] Detailed knowledge of the binding pose can be exploited in drug design to introduce ligand modifications that either position substituents at specific locations within the pocket or modulate binding thermodynamics.

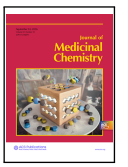
A number of computational protocols have been developed to predict ligand poses by sampling the conformational landscapes of bound ligands using docking[7,8] or more sophisticated simulations.[9−13] Beyond computational predictions, ligand binding behavior is most commonly characterized experimentally using X-ray crystallography,[14−18] cryogenic electron microscopy (cryo-EM),[19−22] or nuclear magnetic resonance (NMR) spectroscopy.[23−29] Of these techniques, NMR is particularly suitable for studying bound ligand poses, as it can be performed in solution and detects all states of the studied system, which means that ensemble quantities are readily accessible.

Among all the macromolecular receptor classes, lectins are particularly well suited for achieving precise control over ligand binding poses due to the high specificity of lectin−carbohydrate recognition. Lectin binding sites are preorganized to expect a specific arrangement of highly directional hydrogen bonds with carbohydrate hydroxyl groups, along with additional CH−π interactions. The high specificity of these interactions underpins the signaling role of lectins in nature, and here we show that it can also be exploited to exert control over the binding behavior of synthetic ligands.

Galectins are a family of lectins that specifically bind β-D-galactosides and are implicated in multiple pathologies, including fibrosis, heart disease, and cancer.[30,31] Galectin-3 (Gal-3) is a prominent member of this family, widely expressed in humans and extensively studied both as a biomarker and as a therapeutic target.[30,32] The galectin binding pocket is divided into binding subsites A−E, originally identified based on the binding of a pentasaccharide motif in galectin-9, but later adopted more universally.[33] Galectins typically recognize galactose as part of a higher oligosaccharide, with the most common binding determinants being disaccharide motifs, such as lactose (Lac) or *N*-acetyllactosamine (LacNAc). Within these canonical disaccharide motifs, the galactose unit is recognized by subsite C of the galectin pocket, while the glucose (or *N*-acetylglucosamine) unit occupies neighboring subsite D.[33,34]

Taking inspiration from this natural binding determinant, we hypothesized that the specificity and preorganization of the

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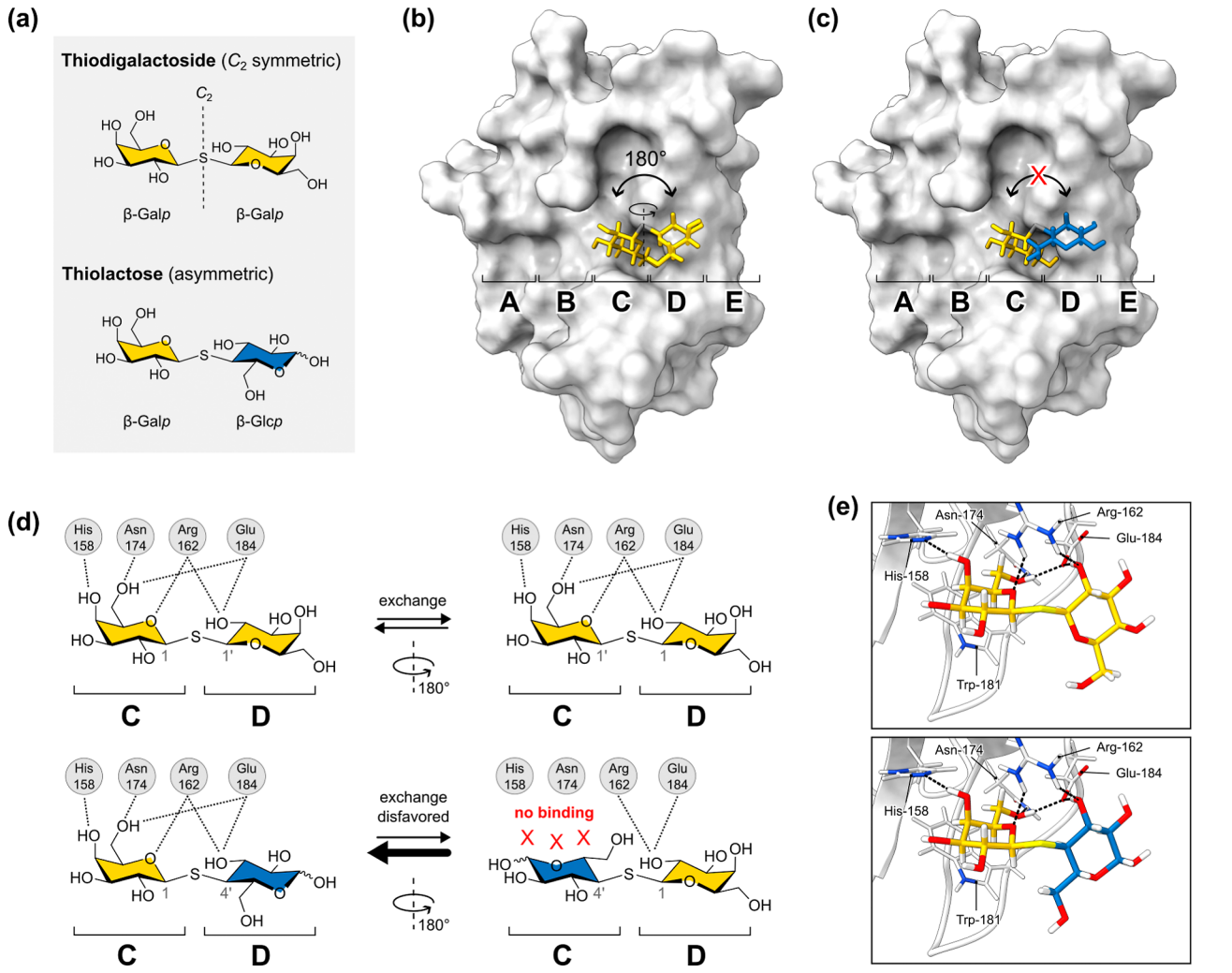


Figure 1. Symmetry-resolved binding of thiodigalactoside and thiolactose to galectin-3. (a) Structures of thiodigalactoside and thiolactose. (b,c) Definition of binding subsites A−E in galectin-3 with bound thiodigalactoside and thiolactose in subsites C and D. (d) Thiodigalactoside can bind symmetrically in two degenerate binding poses by flipping 180°, while thiolactose only binds asymmetrically in a single pose. (e) Structural view of key interactions governing disaccharide recognition in Gal-3.

Gal-3 binding site could be leveraged to achieve precise control over ligand binding behavior through saccharide symmetry. To date, ligand symmetry has been mainly utilized to target C_2- and C_3-symmetric binding sites at the interfaces of homodimeric[35] and homotrimeric[36] receptors. However, symmetry-derived programmatic control over ligand poses in a monomeric binding site has only rarely been achieved[37] and, to our knowledge, has yet to be demonstrated in lectins. To this end, we envisioned two synthetic disaccharide scaffolds—thiodigalactoside and thiolactose—which contain the same binding epitope recognized by Gal-3, yet differ in symmetry (Figure 1a). Thiodigalactoside features a 2-fold rotation symmetry axis (C_2) running through the sulfur atom, meaning that the two symmetrical images of thiodigalactoside can be obtained by rotation about the C_2 axis by 180°. In contrast, thiolactose is asymmetric, and thus the same rotation about the same axis fetches two nonequivalent structures.

As a result, both scaffolds can bind to subsites C and D of Gal-3, but they differ in their ability to flip in the binding site by 180° (Figures 1b,c). When thiodigalactoside is flipped in the pocket, it maintains exactly the same interactions as before, given the two possible orientations in fact represent an identical thermodynamic state. However, when thiolactose is flipped, it loses the ability to bind in subsite C, as the glucose unit is unable to engage in the same interactions as galactose. Accordingly, the flip of the thiolactose scaffold in the binding pocket is disfavored (Figure 1d). For a three-dimensional view of the binding pocket and the key interactions governing recognition of the disaccharide motif in subsites C and D, see Figure 1e.

In this work, we harness this concept to control the bound orientations of small synthetic ligands in Gal-3. We synthesized a series of ligands capable of either symmetric or asymmetric binding and decorated them with fluoroaryltriazole groups, which helped to identify the ligand pose by NMR and showcased the delivery of substituents to specific locations within the pocket. This ligand series serves as a model system for demonstrating how molecular symmetry, in combination with native galectin specificity, can be utilized as a tunable element for controlling binding poses and modulating binding thermodynamics.

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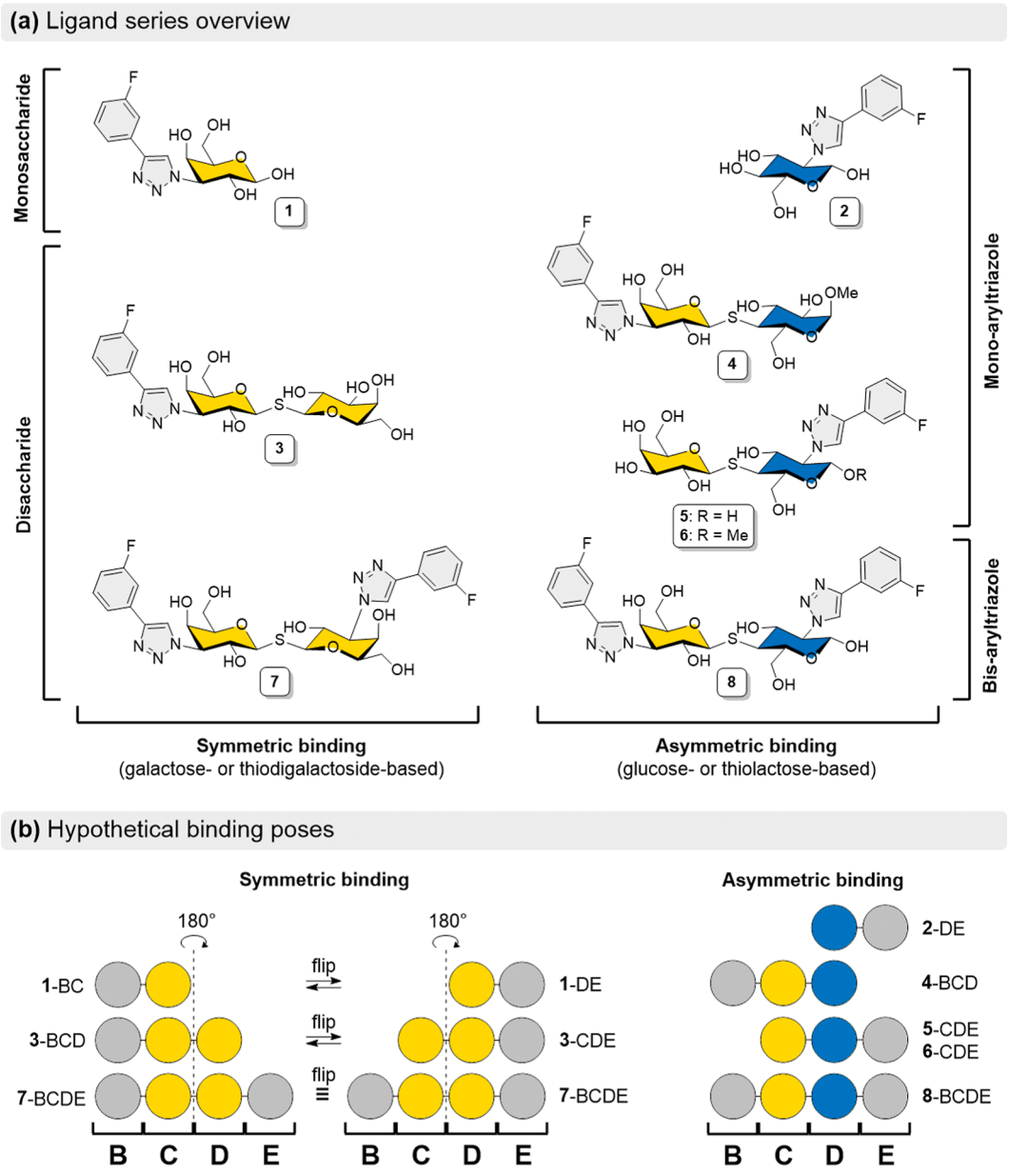


Figure 2. Ligand series overview and synthesis. (a) Overview of the ligand series and ligand categorization according to the number of sugar units, binding symmetry, and number of aryltriazole moieties. (b) Hypothesized symmetry-resolved binding poses, denoted by the ligand number and the subsites occupied by the given pose.

RESULTS AND DISCUSSION

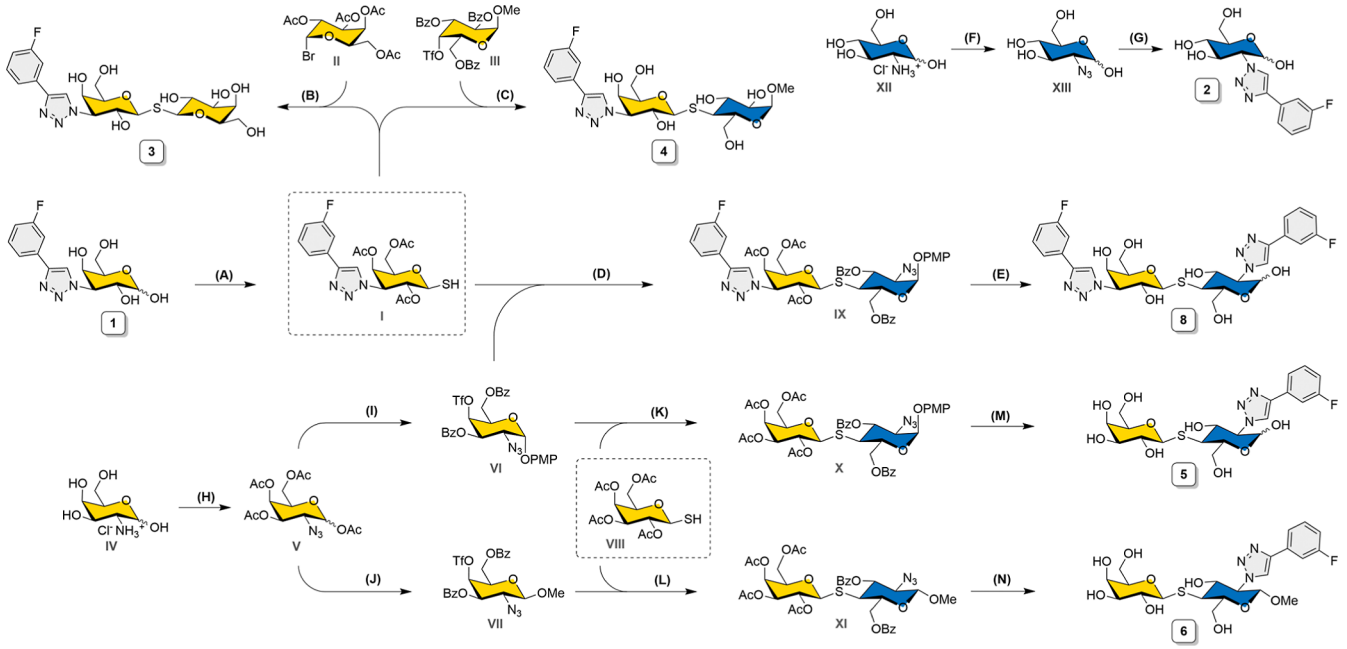
Ligand Series Design and Synthesis

The goal of this study was to investigate the binding preferences of small ligands within the Gal-3 binding pocket based on their symmetry features. For this purpose, we designed a systematic and structurally analogous series of ligands **1**−**8** derived from galactose, glucose, thiodigalactoside, and thiolactose. These ligands were modified in closely analogous positions with a 3-fluorophenyl triazole group, which can be relatively easily installed on the sugar ring, and could serve as an NMR reporter group due to the presence of the ^{19}F nucleus. These aryltriazole groups were placed either at C-3 of the galactose unit or at C-2 of the glucose unit, following established structural motifs of synthetic galectin inhibitors.[38] The resulting ligand series fully covers the chemical and C_2-symmetry space within this group of derivatives, such that each member of the series has an analogous member differing only by the composition of the sugar core or by the presence or absence of the aryltriazole group (Figure 2a). Based on the symmetry of the ligand core, the ligands in this series can hypothetically adopt different poses in the receptor binding site. The galactose- or thiodigalactoside-based ligands **1**, **3**, and **7** can, in principle, bind in two orientations, owing to the ability of the galactose sugar unit to bind in either subsite C or D. As a result, these ligands can flip symmetrically in the binding site and potentially adopt two different poses, that are nonequivalent in the cases of **1** and **3** and equivalent (the same) in the case of **7**. On the other hand, the glucose- or thiolactose-based ligands **2**, **4**, **5**, **6**, and **8** cannot flip in the binding site due to the inability of the glucose unit to bind in subsite C. As a result, these ligands can unequivocally only adopt a single binding pose (Figure 2b).

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Scheme 1. Synthetic Pathways Used for the Preparation of Ligands 1−6 and 8[a]



[a]Reagents and conditions: (A) i. Ac_2O, py, r.t., 3 h, 99%; (ii). TMSOTf, thiourea, ACN, 70 °C, 1 h; (iii). Et_3N, Ac_2O, r.t., 24 h, 73%; (iv). $BnNH_2$, THF, 20 °C, 3 h, 87%. (**B**) (i). K_2CO_3, acetone, r.t., 18 h, 78%; (ii). NaOMe, MeOH, r.t, 3 h, 63%. (**C**) (i). DTT, Et_3N, r.t., 15 h, 60%; (ii). NaOMe, MeOH/CH_2Cl_2 2:1, r.t, 6 h, 84%. (**D**) DTT, Et_3N, r.t., 15 h, 78%. (**E**) (i). NaOMe, MeOH/CH_2Cl_2 1:1, r.t, 2 h, 92%; (ii). CuI, 3-fluorophenylacetylene, DIPEA, DMF, 50 °C, 2 h 83%; (iii). CAN, ACN/H_2O 4:1, 0 °C, 1 h, 51%. (**F**) NfN_3, $NaHCO_3$, $CuSO_4{\cdot}5H_2O$, H_2O/MeOH/Et_2O, r.t, 15 h, 99%. (**G**) CuI, 3-fluorophenylacetylene, DIPEA, DMF, r.t., 2 h 65%. (**H**) (i). NfN_3, $NaHCO_3$, $CuSO_4{\cdot}5H_2O$, H_2O/MeOH/Et_2O, r.t, 15 h, 99%; (ii). Ac_2O, DMAP, py, r.t., 24 h, 92%. (**I**) (i). 4-methoxyphenol, TfOH, 4Å MS, CH_2Cl_2, 0 to 25 °C, 4 h, 98%; (ii). NaOMe, MeOH, r.t., 3 h, 93%; (iii). BzCl, py, 0 °C, 2 h, 74%; iv. Tf_2O, py, CH_2Cl_2, −20 °C, 2 h. (**J**) (i). $TiBr_4$, EtOAc/CH_2Cl_2 9:1, r.t., 72 h; (ii). Ag_2CO_3, MeOH, 60 °C, 2 h, 71%; (iii). NaOMe, MeOH, r.t., 2 h, 69%; (iv). BzCl, py, 0 °C, 3 h, 81%; (v). Tf_2O, py, CH_2Cl_2, −20 °C, 2 h. (**K**) DIPEA, THF, r.t., 15 h, 84%. (**L**) DIPEA, THF, r.t., 72 h, 74%. (**M**) (i). NaOMe, MeOH/CH_2Cl_2 1:1, r.t, 18 h, 61%; (ii). CuI, 3-fluorophenylacetylene, DIPEA, DMF, 50 °C, 1 h, 76%; (iii). CAN, ACN/H_2O 4:1, 0 °C, 1 h, 26%. (**N**) (i). CuI, 3-fluorophenylacetylene, DIPEA, DMF, 50 °C, 2 h; (ii). NaOMe, MeOH, r.t., 72 h, 52%. Abbreviations: CAN: cerium(IV)ammonium nitrate; DIPEA: *N*,*N*-diisopropyl(ethyl)amine; DMAP: 4-dimethylaminopyridine; DMF: dimethylformamide; DTT: 1,4-dithiothreitol; MS: molecular sieves; Nf: nonafluorobutanesulfonyl; PMP: 4-methoxyphenyl; Tf: trifluoromethanesulfonyl; THF: tetrahydrofuran.

The synthetic pathways used to access ligands **1**−**6** and **8** are outlined in Scheme 1. Compound **7** (**GB0139**) was purchased from a commercial supplier and was therefore not synthesized. The galactose derivative **1** was prepared according to our previously published procedure.[39] To obtain the glucose-based ligand **2**, glucosamine hydrochloride (**XII**) was first converted to azide **XIII**, which was subsequently used for a copper-catalyzed click reaction with 3-fluorophenylacetylene to introduce the aryltriazole moiety. To access the disaccharide-based compounds, we utilized compound **1** as the starting material for most of the ensuing syntheses. Using a three-step process, conversion of **1** into its thiouronium derivative and subsequent cleavage by benzylamine afforded anomeric thiol **I**, which served as a glycosyl acceptor for several divergent transformations. Compounds **3** and **4** were both obtained from thiol **I** in two steps via thioglycosylation with bromide **II** and triflate **III**, respectively. In both cases, the reactions proceeded with the configuration inversion at the reaction center, providing the desired configuration of the resulting interglycosidic linkage. To access the key glycosyl donors **VI** and **VII** needed for the synthesis of **5**, **6**, and **8**, galactosamine hydrochloride (**IV**) was first transformed into intermediate azide **V** via diazo transfer,[40] introducing the azido group for subsequent click chemistry. Azide **V** was then subjected to triflic acid-catalyzed glycosylation with para-methoxyphenol (PMP) to give the PMP galactoside, which—after deprotection, selective introduction of benzoyl protecting groups, and triflation—yielded triflate **VI**. Attempts to obtain methyl glycoside **VII** similarly by glycosylation of **V** with methanol were unsuccessful; therefore, **V** was first brominated[41,42] with titanium(IV) bromide, followed by glycosylation with silver(I) carbonate in methanol. Subsequent transformations analogous to those used for compound **VI** afforded triflate **VII** solely as the β-anomer. With the key donor intermediates **VI** and **VII** in hand, a few parallel syntheses were performed to obtain the final compounds **5**, **6**, and **8**. Donor **VI** was first used for thioglycosylation with thiol **I** to give thiolactose derivative **IX**. This intermediate was then used for a copper-catalyzed click reaction with 3-fluorophenylacetylene and deprotected to yield the bis-aryltriazole disaccharide **8**. Alternatively, donors **VI** and **VII** were both reacted in parallel with thiogalactose acceptor **VIII**, affording the thiolactose derivatives **X** and **XI**, respectively. These intermediates were then converted to the final compounds **5** and **6** by click reaction and deprotection in the same manner as for **8**. Full synthetic protocols for all compounds, along with their analytical data, are provided in the Supporting Information.

Saccharide Composition and Substitution Pattern Dictate Ligand Affinities

To investigate the affinities of ligands **1**−**8** and the parent disaccharides thiodigalactoside and thiolactose, we expressed

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Table 1. Affinities of the Investigated Compounds for Gal-3C Determined by Fluorescence Polarization Assay (FPA) and Isothermal Titration Calorimetry (ITC)

	FPA	ITC[a]				
ligand	K_d (μM)	K_d (μM)	ΔG (kcal/mol)	ΔH (kcal/mol)	$-T\Delta S$ (kcal/mol)	N
thiodigalactoside	58 ± 2	31 ± 1	−6.2 ± 0.02	−10.5 ± 0.3	4.3 ± 0.3	1.03 ± 0.02
thiolactose	180 ± 20	86 ± 3	−5.5 ± 0.02	−8.7 ± 0.2	3.1 ± 0.2	1.01 ± 0.02
1	27 ± 3	15 ± 1	−6.6 ± 0.03	−10.4 ± 0.3	3.8 ± 0.4	1.05 ± 0.02
2	>400	n.d.	n.d.	n.d.	n.d.	n.d.
3	0.73 ± 0.04	0.56 ± 0.02	−8.5 ± 0.02	−15.7 ± 0.1	7.2 ± 0.1	1.05 ± 0.06
4	1.0 ± 0.1	1.0 ± 0.1	−8.2 ± 0.1	−13.0 ± 0.1	4.8 ± 0.2	1.08 ± 0.04
5	1.3 ± 0.07	1.2 ± 0.03	−8.1 ± 0.02	−17.4 ± 0.1	9.3 ± 0.1	0.96 ± 0.01
6	1.1 ± 0.08	0.81 ± 0.02	−8.3 ± 0.02	−15.6 ± 0.1	7.3 ± 0.1	1.05 ± 0.01
7	0.011 ± 0.002	0.0074 ± 0.0005	−11.1 ± 0.04	−18.7 ± 0.04	6.3 ± 0.2	0.97 ± 0.01
8	0.013 ± 0.002	0.016 ± 0.0001	−10.6 ± 0.01	−15.8 ± 0.1	5.2 ± 0.1	0.98 ± 0.01

[a]Thermodynamic parameters are reported at 298 K; n.d. = not determined.

the C-terminal domain of human galectin-3 (Gal-3C), which contains the carbohydrate-binding site. Affinities were quantified as dissociation constants (K_d) between the ligands and Gal-3C using two methods: fluorescence polarization assay (FPA) and isothermal titration calorimetry (ITC). FPA is well established for measuring galectin−ligand affinities and, when paired with a suitable mathematical model, enables rapid and accurate determination of K_d values.[43] ITC is considered a gold-standard method[44] for measuring protein−ligand affinities and is able to provide deeper insight into binding thermodynamics by decomposing the binding free energy into enthalpic and entropic components.[45,46] The affinities measured by FPA and ITC demonstrated excellent agreement (Kendall τ = 0.94), confirming the consistency of our data. The results of all affinity measurements are summarized in Table 1.

In general agreement with previously reported affinities of disaccharides toward Gal-3C, thiodigalactoside and thiolactose exhibited K_d values of 58 and 180 μM by FPA, respectively. ITC corroborated these results to the values of 31 and 86 μM. The galactose-based ligand **1** demonstrated an affinity of 27 μM by FPA and 15 μM by ITC, showing a large boost in affinity obtained from the addition of the aryltriazole group. For comparison, unsubstituted methyl galactoside has a much worse reported affinity of approximately 4400 μM by FPA.[47] In contrast, the glucose-based ligand **2** did not produce any measurable FPA signal up to 400 μM, underscoring the importance of the galactose motif for galectin recognition. The monoaryltriazole disaccharides **3**, **4**, **5**, and **6** all exhibited similar affinities within a range of 0.5−1.5 μM by both FPA and ITC, indicating a comparably strong effect of the aryltriazole moiety when placed on the galactose or glucose unit. Finally, the bis-aryltriazole compounds **7** and **8** exhibited almost identical low-nanomolar affinities for both methods, demonstrating the added benefit of the second aryltriazole group in each case.

The ITC experiments were also used to obtain more detailed information about the binding thermodynamics of the ligands (Table 1). The binding stoichiometry (N) was close to unity for all tested compounds, confirming that each ligand interacts with a single site on the protein in a 1:1 manner. Titrations of two randomly selected ligands (**3** and **4**) performed in buffers with different enthalpies of ionization (Tris−HCl or HEPES) yielded the same binding enthalpies, indicating that no net proton transfer occurs during binding under our experimental conditions. As expected, binding of all tested ligands was driven by a favorable enthalpic contribution and opposed by an unfavorable entropic penalty, although the specifics of the enthalpy−entropy compensation varied among ligands. Between the two bare disaccharide scaffolds, thiodigalactoside shows three-times stronger binding than thiolactose, owing to more favorable enthalpy (−10.5 compared to −8.7 kcal/mol, $\Delta\Delta H$ = −1.8 kcal/mol), which is not fully compensated entropically ($-T\Delta\Delta S$ = 1.2 kcal/mol). Within the monoaryltriazole ligands (**3**−**6**), compound **4** stood out by displaying a relatively less favorable enthalpy but a smaller entropic penalty than **3**, **5**, and **6**, thereby achieving its affinity through a smaller entropy loss. Between the bis-aryltriazole ligands **7** and **8**, the thiodigalactoside-based **7** exhibited somewhat more enthalpy-driven binding than the thiolactose-based **8**. However, such small differences in enthalpy−entropy compensation obtained from ITC should be interpreted with caution.[46]

Collectively, these measurements enabled us to map the affinity trends within the ligand series using two independent methods and to determine the binding thermodynamics of the more potent ligands.

^{19}F NMR Reveals Symmetry-Resolved Ligand Binding

To investigate the binding of ligands **1**−**8** to Gal-3C, we took advantage of the fluorine atom preinstalled in all of the studied molecules. The ^{19}F nucleus is ideal for this investigation due to its high abundance, high sensitivity, and wide range of chemical shift.[48] Titrations using ^{19}F NMR were used previously to investigate the subsite preferences of compounds **3** and **7** in galectins, although we note that erroneous assignment of the free and bound signals was used.[49] We later adapted similar measurements for ferrocene- and ruthenium-based glycomimetics[41] in galectin-1 and -3C, and more recently to elucidate the binding of a novel class of *N*-aryl-*N*-lactosylamide inhibitors[50] to Gal-3C.

The ligand series in the present study is intentionally designed so that the core saccharide structures are modified with one or two aryltriazole groups, which are used to deliver the ^{19}F labels to either subsites B or E, where they could be distinguished by NMR (Figure 3a). The ^{19}F NMR titrations of compounds **1**−**8** to Gal-3C revealed clear evidence of symmetry-resolved binding, manifesting in the appearance of distinct signals corresponding to the localization of the ^{19}F atom in different regions of the receptor. These signals could easily be discerned by their chemical shifts and assigned to

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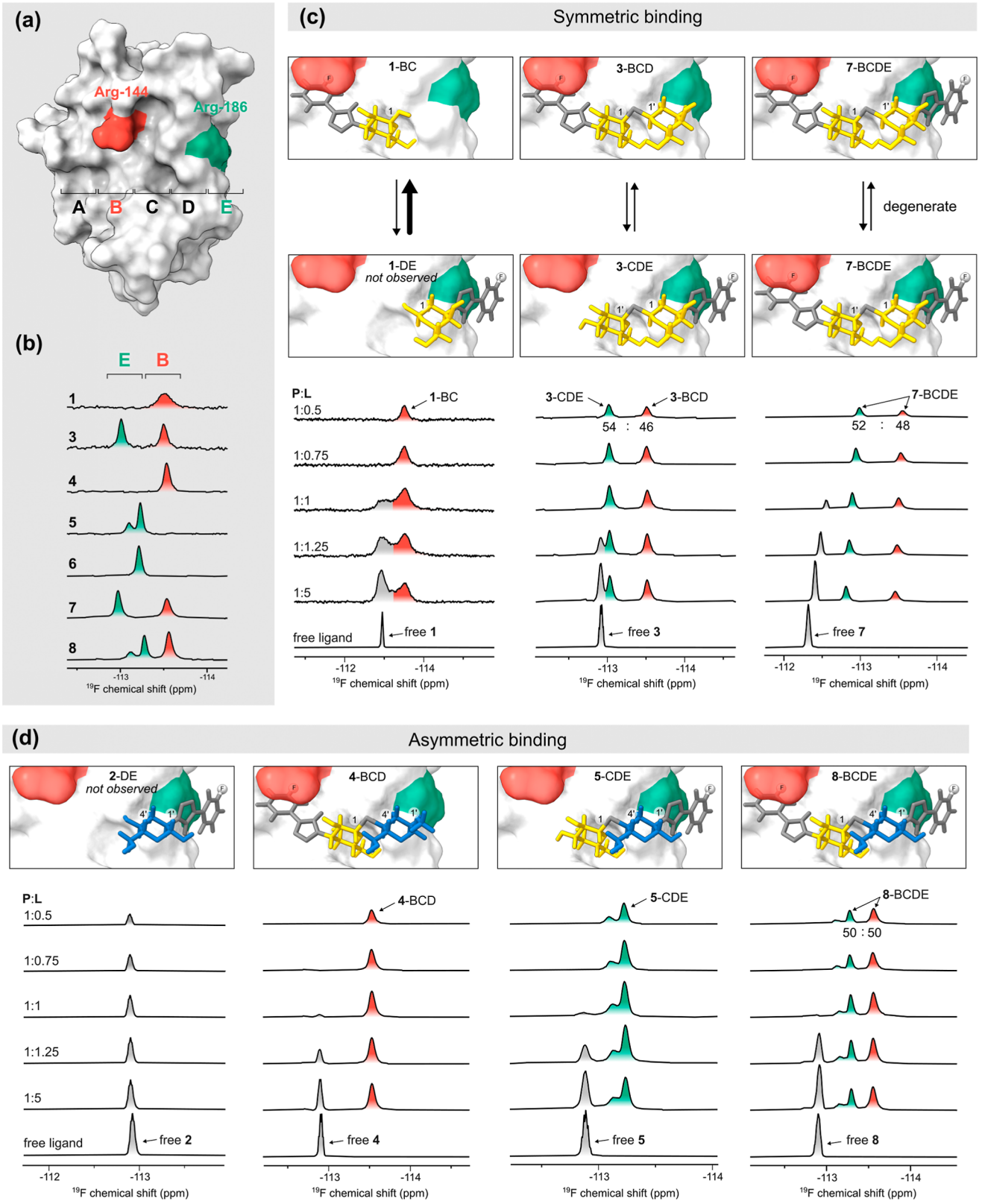


Figure 3. Ligand-observed ^{19}F NMR spectroscopy of ligands **1**−**8**. (a) Definition of subsites B and E, which are responsible for the binding of the aryltriazole group. The key residues in these subsites (Arg-144 and Arg-186, respectively) are highlighted. (b) Overview of the detected bound states at a [P]:[L] concentration ratio of 1:0.5. Ligand **2** is omitted since no bound state was observed. (c) Symmetric binding of ligands **1**, **3**, and **7**. (d) Asymmetric binding of ligands **2**, **4**, **5**, and **8**. Ligand **6** is omitted for simplicity.

either subsite B (approximately −113.6 ppm) or subsite E (ranging between −112.9 and −113.4 ppm) (Figure 3b).

For the thiodigalactoside-based compounds **3** and **7**, which can bind symmetrically in two possible orientations, two bound signals were indeed observed (Figure 3c). In the case of

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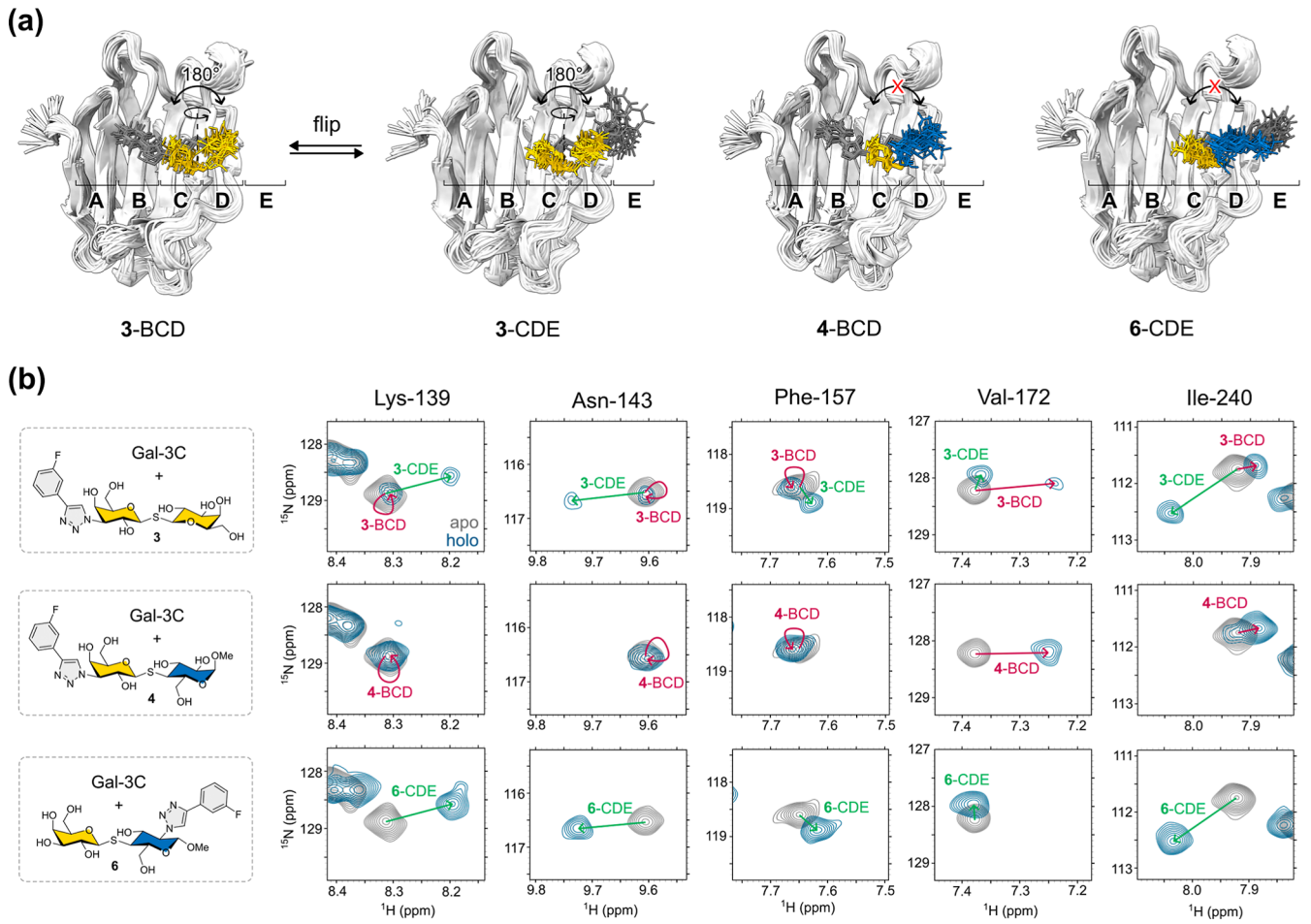


Figure 4. Protein-observed NMR spectroscopy of monoaryltriazole ligands **3**, **4**, and **6**. (a) Hypothesized binding poses of compounds **3**, **4**, and **6**. While compound **3** binds symmetrically in an equilibrium of poses **3**-BCD and **3**-CDE, compounds **4** and **6** bind asymmetrically in poses **4**-BCD and **6**-CDE, respectively. (b) Comparison of selected cross-peaks from ^{1}H−^{15}N HSQC NMR spectra of compounds **3**, **4**, and **6**, showing the symmetry-resolved binding behavior.

compound **3**, these two signals correspond to two nonequivalent thermodynamic states (**3**-BCD and **3**-CDE), where the aryltriazole group occupies either subsite B or E. In the case of **7** on the other hand, the two signals correspond to a single degenerate binding pose (**7**-BCDE), where both subsites B and E are simultaneously occupied by both aryltriazole groups. For both compounds **3** and **7**, an approximate 1:1 ratio of integral intensities between the two bound signals was observed. This ratio is fully expected for compound **7** and supports its single pose. For compound **3**, the equimolar ratio indicates the nearly equal probabilities of poses **3**-BCD and **3**-CDE. For compound **1**, two binding poses (**1**-BC and **1**-DE) are theoretically possible, yet only one bound signal was observed, corresponding to the **1**-BC pose. This result clearly shows that the **1**-BC pose is thermodynamically preferred over **1**-DE, an observation that is readily explained by stronger binding of galactose in subsite C over subsite D.

Complementary evidence was obtained for compounds **2**, **4**, **5**, **6**, and **8**, which bind asymmetrically by design (Figure 3d). For the glucose-based compound **2**, where only the **2**-DE pose is theoretically possible, no bound signal was observed at all. This finding tallies with the observation made for compound **1**, where the **1**-DE pose was also too weak to detect. These results together indicate unfavorable binding energy of poses where subsite C is not occupied by the canonical galactose motif. The thiolactose-based compounds **4**, **5**, **6**, and **8** can only adopt a single binding pose, enforced by the asymmetry of the thiolactose core. The monoaryltriazole compounds **4**, **5**, and **6** differ in the position of the aryltriazole group—and consequently in the location of the fluorine atom—which is clearly reflected in the ^{19}F NMR spectra of their bound states. Compound **4** exhibits a single bound signal corresponding to the **4**-BCD binding pose, where the fluorine atom resides in subsite B. In contrast, compounds **5** and **6** each show a single major bound signal corresponding to poses **5**-CDE and **6**-CDE, respectively, where the fluorine resides in subsite E. Importantly, these findings demonstrate selective, asymmetric delivery of the aryltriazole substituent and, in principle, other substituents to either subsite B or E, depending on the substitution pattern of the ligand. Interestingly, we observed an additional minor peak in the bound spectrum of **5** compared to **6** (see comparison in Figure 3b). This difference is likely due to compound **5** being a free anomer (ratio in the unbound state is α/β 1:2 in MeOD), whereas compound **6** is purely a β-glycoside. This likely results in an additional bound conformation of the aryltriazole group in the α-anomer of **5**, leading to an additional signal in subsite E. Finally, the bis-aryltriazole compound **8** also binds in only a single pose (**8**-BCDE), reflected in its bound spectrum by two peaks with 1:1 integral intensities corresponding to the two nonequivalent

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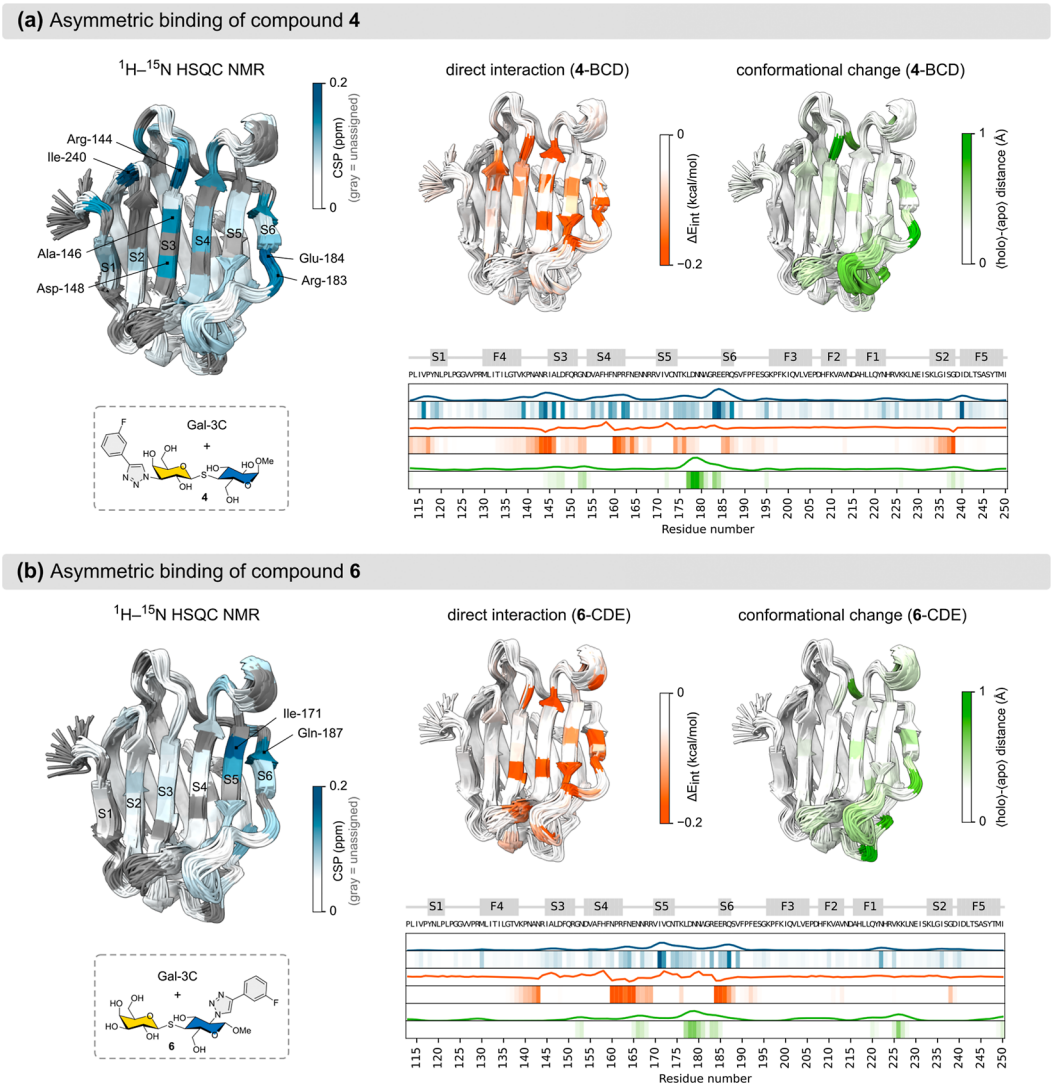


Figure 5. Chemical shift perturbation signatures induced by **4** and **6**. (a, b) Asymmetric binding of thiolactose-based compounds **4** and **6** elucidated by protein-observed NMR and in silico analysis. The displayed quantities are mapped on an ensemble of 50 structures obtained from REST simulations. Ligands are omitted from the structural ensembles for clarity.

positions of the two fluorine atoms in subsites B and E. Like compound **5**, compound **8** also displays a minor secondary bound peak in subsite E, which likely corresponds to its α-anomer.

Taken together, our ^{19}F NMR experiments illustrate the conformational control imposed within the ligand series by the symmetry features of the saccharide cores, allowing us to decipher the individual symmetry-resolved poses of all the studied ligands.

Ligand Poses Induce Distinct Protein-Observed NMR Signatures

To further validate our symmetry-resolved binding hypothesis beyond the information obtained from the ^{19}F NMR measurements, we examined the binding behavior of the key monoaryltriazole compounds **3**, **4**, and **6** using protein-observed NMR spectroscopy. These compounds were selected to represent both symmetric binding (compound **3**) and asymmetric binding with the aryltriazole group in both possible positions (compounds **4** and **6**) (Figure 4a). All three ligands have K_d values between 0.5 and 1.5 μM, resulting in a slow exchange between bound and free states relative to the NMR time scale (k_{off} is much smaller than the chemical-shift difference).[51,52] As a result, titration of the ligands to ^{15}N, ^{13}C-labeled Gal-3C led to a gradual disappearance of apo signals and the concurrent appearance of holo signals without a smooth transition through intermediate states, making assignment of the bound states more difficult.

A comparison of the apo and holo ^{1}H–^{15}N HSQC spectra of compounds **3**, **4**, and **6** for selected residues is shown in Figure 4b. The compound **3**, which is hypothesized to bind in an equilibrium of two different poses (**3**-BCD and **3**-CDE), indeed demonstrated two sets of cross-peaks in the holo spectra, confirming the presence of the two bound states of compound **3**. While it is not possible to ascertain which set of cross-peaks belongs to which bound pose directly from the spectra of **3**, it could be deduced indirectly from the comparison with the holo spectra of the thiolactose-based ligands **4** and **6**. Both of these compounds are assumed to bind in a single pose (**4**-BCD and **6**-CDE, respectively), consistent with the observation of a single set of cross-peaks for each compound. In addition, the two sets of signals observed for compound **3** closely matched those observed for **4** and **6**,

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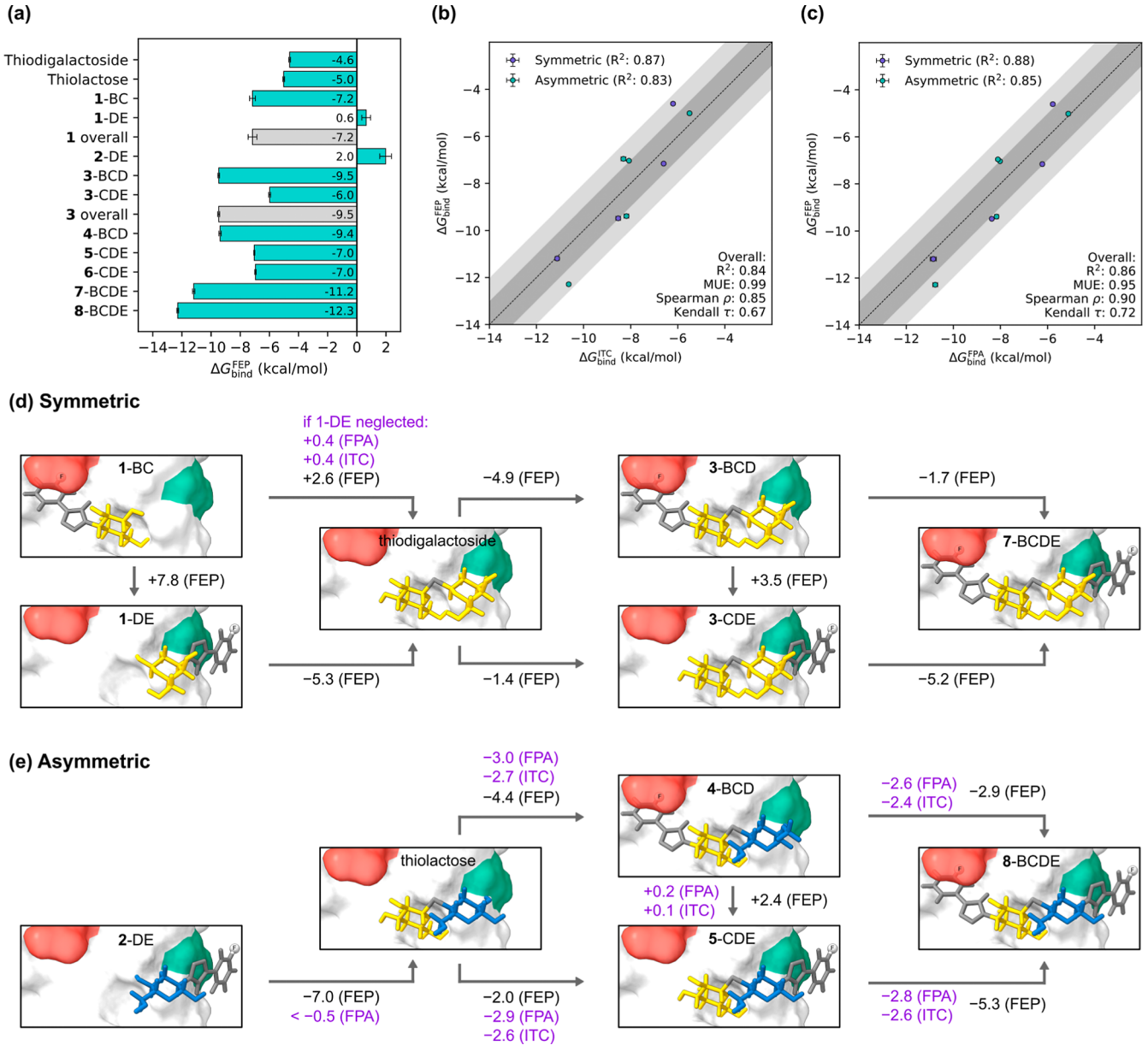


Figure 6. Alchemical free energy calculations. (a) Binding free energies of individual binding poses calculated by FEP. (b,c) Correlation between binding free energies calculated by FEP and experimental binding free energies obtained from ITC and FPA. The darker and lighter gray colors indicate the errors of 1 and 2 kcal/mol. (d,e) Maps of ligand poses in both the symmetric and asymmetric series with their relative binding free energies $\Delta\Delta G^{FEP}_{bind}$ obtained from maximum likelihood estimate from FEP calculations (black). Experimental $\Delta\Delta G_{bind}$ values are also included where deducible (purple). All energies are in kcal/mol.

further supporting the symmetry-resolved binding model, where the binding behavior of **3** is essentially a combination of the bound poses adopted by **4** and **6**. This manifests in the cross peaks observed in the holo spectra of **3** being a superset of those observed in the holo spectra of **4** and **6**.

To further investigate the specifics of the protein−ligand interactions in our model system, we determined the chemical shift perturbation (CSP) of the Gal-3C backbone induced by compounds **4** and **6** and mapped it onto the conformational ensembles obtained from in silico simulations (Figure 5). It is generally accepted that CSP can arise either from direct ligand interactions with residues in the binding site or from ligand-induced conformational perturbations of the receptor (allostery).[52,53] To distinguish between these two effects, we extracted two proxy quantities from the simulations and used them as qualitative descriptors of the ligand binding spatial patterns.

First, we calculated a per-atom decomposition of the gas-phase interaction energy (ΔE_{int}) derived from force field. This quantity should be interpreted strictly as a qualitative spatial descriptor and does not include the full thermodynamic contributions to binding, such as desolvation, configurational entropy, structural reorganization, and the probabilities of alternative states. Second, we estimated receptor conformational rearrangements by calculating the Euclidean distance between the average position of each atom in the holo and apo trajectories. Both of these descriptors were compared with the experimental CSP maps to assess whether the perturbed receptor regions spatially coincide with the descriptor patterns observed in the simulations.

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The CSP signatures of compounds **4** and **6** (Figures 5a,b, respectively) enabled us to examine the asymmetric binding of the aryltriazole group in subsites B and E as two distinct, isolated cases. At first glance, it is immediately apparent that compound **4** induces a more extensive CSP signature, affecting residues Arg-144, Ala-146, Asp-148, Arg-183, Glu-184, and Ile-240, corresponding to subsite B and its vicinity. In contrast, compound **6** induces a more localized CSP pattern involving mostly the residues Ile-171 and Gln-187, localized in subsite E. In both cases, the perturbed regions of the receptor generally agree with the expected localization of the aryltriazole group, supporting our assumptions about the orientations of both ligands within the receptor.

Intriguingly, the observed CSP patterns were in part correlated with the proxy quantities taken from the simulations. For example, the large CSP on sheets S2, S3, and S4 induced by compound **4** generally correlates with the per-atom interaction energy decomposition calculated for the **4**-BCD pose. Conversely, the CSP on the S6 sheet observed for the compound **6** agrees with the interaction energy pattern detected in the simulation of **6**-CDE. Further insight was gleaned from Arg-144, a key residue in subsite B perturbed in the spectra of compound **4**. This residue displays both a very strong direct interaction and conformational perturbation in the simulation of **4**-BCD, but a much smaller direct interaction and no conformational perturbation in **6**-CDE. These findings further support the assignment of compound **4** to the **4**-BCD pose.

From a structural standpoint, our results indicate that the CSP observed on the sheets of Gal-3C may likely be attributed to direct effects, as these regions of the protein are relatively rigid. In contrast, the loop regions exhibit greater flexibility and are likely to undergo stronger conformational perturbations upon ligand binding. However, this could not be directly confirmed by our experiments due to the invisibility of the most flexible loop regions in the NMR spectra.

All in all, the protein-observed NMR experiments corroborated the symmetry-resolved pose assignments of compounds **4** and **6** and illustrate that the CSP signatures associated with their binding to Gal-3C are largely due to the hypothesized binding poses **4**-BCD and **6**-CDE. These findings confirm the asymmetric control of the ligand poses as well as the selective delivery of the aryltriazole group to either subsite B or E, depending on its position.

Free Energy Calculations Dissect the Thermodynamics of Individual Ligand Poses

One of the important questions in this study was to discern the affinities of the individual binding poses. This task is essentially impossible to solve experimentally, as experimental approaches typically only provide free energy changes averaged across the entire thermodynamic ensemble. Nevertheless, alchemical free energy perturbation (FEP) calculations are, in theory, well-positioned to answer this question, as they can estimate the binding free energies of individual binding poses.[54−58] We calculated binding free energies (denoted here as $\Delta G_{\mathrm{bind}}^{\mathrm{FEP}}$) for all hypothesized binding poses using the OpenFE software package[59] (Figure 6a). For ligands with two nonequivalent binding poses, the overall binding free energy was calculated by Boltzmann-weighting the individual pose-specific binding free energies and combining them into a single effective binding free energy. After referencing the experimental ITC data, the calculated $\Delta G_{\mathrm{bind}}^{\mathrm{FEP}}$ values closely correlated with the experimental affinities obtained from both ITC (R^2 = 0.84; Figure 6b) and FPA (R^2 = 0.86; Figure 6c), corroborating the validity of our computational approach. These results enabled us to construct a diagram of the individual ligand poses in both the symmetric and asymmetric series and the relative binding free energies between them (Figure 6d).

Indeed, the calculated $\Delta G_{\mathrm{bind}}^{\mathrm{FEP}}$ values of the individual binding poses reproduced the experimental affinity trend fairly well, supporting the overall validity of the hypothesized poses. For the galactose-based compound **1**, the **1**-BC pose was clearly identified as the dominant pose over **1**-DE by 7.8 kcal/mol, in agreement with the ^{19}F NMR titration results. Also consistent with our experimental data, the **2**-DE pose, predicted as the only possible pose for compound **2**, showed a similarly poor $\Delta G_{\mathrm{bind}}^{\mathrm{FEP}}$ as **1**-DE, which is consistent with a poor experimental affinity measured for compound **2**.

For compound **3**, the **3**-BCD pose was identified by FEP as more favorable than **3**-CDE by 3.5 kcal/mol. This difference in $\Delta G_{\mathrm{bind}}^{\mathrm{FEP}}$ contrasts with the experimental results for both ligand-observed ^{19}F NMR and protein-observed NMR spectra, both of which yielded an approximate 1:1 ratio for the two bound states of compound **3**. For the thiolactose-based monoaryltriazole compounds **4**, **5**, and **6**, the **4**-BCD pose also exhibited more favorable $\Delta G_{\mathrm{bind}}^{\mathrm{FEP}}$ than **5**-CDE and **6**-CDE, again pointing to a preferred orientation of the aryltriazole group in the B subsite estimated by FEP, contrasting with spectroscopy and affinity measurements by FPA and ITC. The disagreement between FEP estimates and experimental results for compounds **3** and **4** appears to be a clear limitation of the present computational model. Although both NMR and in vitro affinity experiments can also contain systematic errors, the agreement between independent NMR measurements and both FPA and ITC affinity assays for **3** and **4** strongly suggests that our FEP calculations are overestimating the preference for the B-subsite orientation in these monoaryltriazole compounds. The source of this error is hard to determine with certainty in this study, but may include higher flexibility of the aryltriazole group in subsite E compared to B, water-mediated interactions, or other slow degrees of freedom that are hard to converge despite extensive sampling. Additionally, force-field inaccuracies, partial-charge assignment, or systematic errors introduced by the RBFE network might be at play. Finally, the excellent experimental affinities of the bis-aryltriazole compounds **7** and **8** were reproduced by the FEP calculations of **7**-BCDE and **8**-BCDE poses, displaying much better binding free energy than any of the monoaryltriazole ligands and confirming the meaningfulness of the bis-substituted BCDE binding motif.

Together, the calculations presented here offer microscopic insight into the binding thermodynamics of individual ligand poses simulated as isolated states and quantify their binding free energies.

Entropy Loss in Symmetric and Asymmetric Binding

For compounds **1** and **3**, where two nonequivalent binding poses can be populated, it is possible to estimate the configurational free energy contribution arising from the coexistence of these two binding configurations. This free energy contribution arises from the larger phase space accessed by the symmetrically binding ligands in the bound state and determines the amount of entropic binding penalty. In the edge case of two poses with identical binding free energies, where the entropy bonus is most favorable, its theoretical maximal value would amount to $\Delta\Delta G_{\mathrm{config}} = -RT \ln(2)$,

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corresponding to −0.41 kcal/mol at 298 K. Using populations of the two bound states of compound **3** obtained from the integration of its bound ^{19}F NMR spectra, $\Delta\Delta G_{config}$ of **3** was calculated to −0.37 kcal/mol at 298 K. The NMR-derived near-equal populations of compound **3** therefore imply a configurational contribution close to the upper theoretical limit. In contrast, the FEP-derived estimate is much smaller because FEP predicts the **3**-BCD pose to dominate. Specifically, $\Delta\Delta G_{config}$ for compound **3** was estimated from FEP to be -1.6×10^{-3} kcal/mol; however, as discussed earlier, we expect this value is erroneously small due to FEP overestimating the population of **3**-BCD compared to **3**-CDE. For compound **1**, $\Delta\Delta G_{config}$ is not available from NMR, because the **1**-DE pose was not even observed, and FEP-derived estimate is calculated to -1.0×10^{-6} kcal/mol. Collectively, these results suggest a negligible role of $\Delta\Delta G_{config}$ for compound **1**.

The NMR-derived estimates of $\Delta\Delta G_{config}$ for compound **3** are consistent with the differences in experimental affinities measured for the monoaryltriazole compounds **3**, **4**, **5**, and **6**. The experimental dissociation constant of the symmetrically binding compound **3** is 0.73 μM by FPA and 0.56 μM by ITC, while the asymmetrically binding compounds **4**, **5**, **6** all display somewhat weaker affinity (1.0, 1.3, 1.1 μM by FPA and 1.0, 1.2, 0.81 μM by ITC, respectively), as shown in Table 1. These differences in dissociation constants correspond to 0.20−0.45 kcal/mol, consistent with the estimated value of $\Delta\Delta G_{config}$ for compound **3**. However, this trend is not clearly corroborated by the $-T\Delta S$ contributions obtained from ITC, suggesting that other sources of entropy dominate this term, making direct experimental measurement of $\Delta\Delta G_{config}$ impossible.

While the exact values of the entropy contribution are hard to obtain due to inherent limitations of both experimental and computational approaches, the magnitude of this contribution seems to be relatively small compared with the overall affinity differences across the ligand series, suggesting that loss of the two-pose configurational contribution is unlikely to be a major practical penalty for the asymmetric thiolactose-based design.

CONCLUSIONS

In this work, we explore the use of axial symmetry, in conjunction with the native specificity of lectin−carbohydrate recognition, as a rational principle for controlling ligand poses in Gal-3. We devised a highly analogical series of Gal-3 ligands based on galactose, glucose, thiodigalactoside, and thiolactose scaffolds, strategically equipped with aryltriazole substituents bearing a ^{19}F label. By combining ligand-observed and protein-observed NMR spectroscopy with in-silico modeling, we identified symmetry-resolved binding behavior within this series, where ligands adopt different poses based on their global C_2 symmetrical features. These results demonstrate programmatic control over the binding poses and pave the way for the design of a new generation of disaccharide-based Gal-3 ligands with predictable binding behavior.

Specifically, our results confirm the key role of galactose for binding subsite C of Gal-3, and interchangeable roles of galactose and glucose for binding subsite D. The nearly equal probabilities of poses **3**-BCD and **3**-CDE observed by both ligand- and protein-observed NMR for compound **3** point to a similar affinity of the 3-fluorophenyl triazole group for subsites B and E. This is also evidenced by the nearly identical affinity of compounds **4**, **5**, and **6**, as measured by both ITC and FPA in vitro assays. Under this assumption, it could also be deduced that binding of galactose is much stronger in subsite C over subsite D, because only the **1**-BC pose, but not the **1**-DE pose, was observed for compound **1**, leaving this canonical motif as a strong design choice.

Importantly, the symmetry-resolved binding model proposed here demonstrates that the asymmetric thiolactose-based compounds **4**, **5**, **6**, and **8** could carry the 3-fluorophenyl triazole group (and theoretically any other group) specifically into either subsite B or E, depending on the thiolactose substitution. Such substituent delivery was previously not possible with the commonly used thiodigalactoside scaffold, which can only target both subsites B or E at once due to its symmetry. These findings open new opportunities for more targeted design of the peripheral modifications of galectin inhibitors, aiming at specific locations within the binding site. Given the relatively low affinity of unmodified saccharides toward Gal-3 and hence the need for additional modifications to achieve drug-like potency, such design opportunities are valuable for further development of galectin-targeting therapeutics and imaging agents. Overall, our study provides a guide for the delivery of substituents specifically into anticipated regions of the receptor pocket by derivatization of the galactose, thiodigalactoside, or thiolactose scaffolds at defined positions. These findings can be extended to various other substituents and potentially even different derivatization hotspots.

To dissect the thermodynamics of the individual poses, we calculated the binding free energies of all hypothesized poses using alchemical FEP, which further extended our results beyond the ensemble-averaged experimental observations. This dissection allowed us to map the free energy differences between different ligand orientations and quantify the binding free energy contributions of the individual ligand building blocks in different receptor regions. We also used our spectroscopic and theoretical results to estimate idealized configurational free energy contributions in the bound state for compounds **1** and **3**, which bind in two nonequivalent configurations. In both cases, these entropy contributions were found to likely be below approximately 0.4 kcal/mol, while it is not clear whether the differences in experimental affinities between compounds with symmetric and asymmetric binding can solely be attributed to this factor. This suggests that the loss of configurational entropy arising from the asymmetric binding is unlikely to bring a major practical hindrance for designing future thiolactose-based inhibitors or other asymmetrically binding galectin inhibitors.

EXPERIMENTAL SECTION

Synthesis

All reagents and solvents were purchased from commercial suppliers and used without further purification unless otherwise stated. Anhydrous CH_2Cl_2 and THF were obtained by distillation from CaH_2 and $CaH_2/LiAlH_4$, respectively, under an argon atmosphere. Thin-layer chromatography (TLC) was performed on silica gel 60 F_{254} (Sigma-Aldrich). Spots were visualized under UV light (254 nm), by staining with aqueous $H_2SO_4/CeSO_4$, or using a combination of both methods. Column chromatography was performed on silica gel 60 (70−230 mesh, Material Harvest). Unless otherwise noted, solutions were concentrated under reduced pressure at temperatures below 40 °C. Final compounds were purified by reversed-phase HPLC using a C8 or C18 column. Analytical HPLC was conducted on a Waters Alliance 2695 system with UV detection (Waters 2996) using a Kromasil 5 μm Polar C18 LC column (100 Å, 250 × 4.6 mm) with gradient elution (10% to 90% CH_3OH in 20 min, then 5 min

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isocratic 90% CH_3OH, flow rate = 0.7 mL/min). All compounds are >95% pure by HPLC analysis. The HPLC traces for all synthesized final compounds are included in the Supporting Information.

All ^1H and ^{13}C NMR spectra were recorded on the following spectrometers: Bruker Avance III HD 400 MHz (with CryoProbe Prodigy), Bruker AVANCE III HD 500 MHz, Bruker AVANCE III HD 600 MHz, JEOL ECZR 400 MHz, and JEOL JNM-ECZR 500 MHz. ^1H and ^{13}C resonances were fully assigned using ^{13}C-APT, $^1H-^1H$ COSY, $^1H-^{13}C$ HSQC, $^1H-^{13}C$ HMBC, and NOESY techniques. Chemical shifts were referenced to the residual solvent signal or to internal standards (1,4-dioxane or *tert*-butanol). Coupling constants (J) are reported in Hz with the following splitting abbreviations: s = singlet, d = doublet, t = triplet, q = quartet, m = multiplet, bs = broad signal. Infrared (IR) spectra were recorded on a Nicolet 6700 spectrometer (Thermo Scientific). Absorption maxima (ν_{max}) are reported in wavenumbers (cm^{-1}). High-resolution mass spectra (HRMS) were measured on an LTQ Orbitrap XL spectrometer (Thermo Scientific) using electrospray ionization (ESI). Nominal and exact m/z values are reported in Daltons. Optical rotations were measured on an AUTOPOL IV polarimeter (Rudolph Research Analytical, USA) at 20 °C using a sodium D-line (λ = 589 nm) and a path length (l) of 1.0 dm. Concentrations (c) are given in g per 100 mL. Specific rotation values are reported as dimensionless numbers with implied units of $deg{\cdot}mL{\cdot}g^{-1}{\cdot}dm^{-1}$. Melting points (mp) were recorded on an MPM-HV3 capillary melting-point apparatus (Schorpp-Gerätetechnik, Germany) and are reported uncorrected in degrees Celsius (°C).

Protein Preparation

The human galectin-3 C-terminal domain (Gal-3C) was cloned and expressed as previously described.[41] For the preparation of uniformly ^{13}C- and ^{15}N-labeled Gal-3C, *Escherichia coli* BL21 (DE3) cells, transformed with the appropriate plasmid, were pregrown in 5 mL of medium A (medium A per liter: 100 mL M9 medium (10×), 5 mL trace elements solution (200×), 2 g ^{13}C-glucose, 0.5 g $^{15}NH_4Cl$, 1 mL 1 M $MgSO_4$, 0.3 mL 1 M $CaCl_2$, 1 mL biotin (1 mg/mL stock), 1 mL thiamin (1 mg/mL stock), 100 mg ampicillin; 10 × M9 medium per liter: 67.8 g Na_2HPO_4, 30 g KH_2PO_4, 5 g NaCl; 200 × trace element solution per 500 mL: 5 g EDTA, 0.83 g $FeCl_3{\cdot}6H_2O$, 84 mg $ZnCl_2$, 13 mg $CuCl_2{\cdot}2H_2O$, 10 mg $CoCl_2{\cdot}6H_2O$, 10 mg H_3BO_3, 1.6 mg $MnCl_2{\cdot}6H_2O$ and 10 mg Na_2MoO_4). Cultures were incubated overnight at 37 °C with shaking (180 rpm). The overnight culture was transferred into 1 L of medium A and grown at 37 °C with shaking (180 rpm) until the OD600 reached a value of between 0.8 and 1.0. Protein expression was induced by the addition of IPTG to a final concentration of 0.4 mM together with a temperature shift to 22 °C. The culture was cultivated for 16 h under these conditions and then spun down for 10 min at 8000*g*. The cell pellet from 1 L of media was resuspended in 10 mL of lysis buffer (100 mM Tris, pH 7.4, 0.5 M NaCl, 5 mM β-mercaptoethanol, 5 mM $MgCl_2$, 0.5 mg/mL lysozyme and 125 U Novagen Benzonase) and sonicated for 2 min (MSE Soniprep 150; 75% power, amplitude 14 μm). The lysate was centrifuged at 20,000*g* for 20 min. The cleared lysate was run through a chromatographic column containing an appropriate amount of lactosyl−sepharose (binding capacity of approximately 10 mg/mL), prepared as previously described.[60] The column was washed with 10 column volumes of wash buffer (100 mM Tris, pH 7.4, 0.5 M NaCl, and 5 mM β-mercaptoethanol) and galectin was eluted using an appropriate amount of elution buffer (50 mM Tris, pH 7.4, 150 mM NaCl, 5 mM β-mercaptoethanol, and 0.1 M β-Lactose). The eluted protein was concentrated (Amicon Ultra-15, Ultracel 10K; Merck Millipore), dialyzed twice (1:200) against PBS containing 5 mM β-mercaptoethanol using D-tube Dialyzer Mega (MWCO 3.5 kDa; Merck Millipore), and stored at 4 °C.

Fluorescence Polarization Assay

Fluorescence polarization measurements were performed as previously described.[50] Briefly, the fluorescent probe 3-deoxy-3-[4-(fluorescein-5-yl-carbonylaminomethyl)-1*H*-1,2,3-triazol-1-yl]-β-D-galactopyranosyl-3-deoxy-3-[4-(3-fluorophenyl)-1*H*-1,2,3-triazol-1-yl]-1-thio-β-D-galactopyranoside was used.[41,61] The dissociation constant of the probe was determined by direct binding experiments, in which the Gal-3C protein was titrated to a fixed probe concentration (5 nM). Data from direct binding experiments were fitted using a quadratic model accounting for receptor depletion. Dissociation constants of the ligands were determined by competitive binding experiments using 50 nM of Gal-3C and 5 nM of the probe. Data obtained from competitive binding experiments were fitted using a cubic model described by Wang.[62] All inhibitor dissociation constants are reported as mean ± SD from three independent experiments, each performed in triplicate. Nonlinear regression analyses of both direct and competitive binding experimental data were performed using BindCurve.[63]

Isothermal Titration Calorimetry

The binding of ligands to Gal-3C was monitored at 25 °C using a VP-ITC microcalorimeter (MicroCal Inc./Malvern Instruments Ltd., UK). Reactant solutions were prepared in 25 mM Tris−HCl (pH 8.0), containing 300 mM NaCl, 5 mM 2-mercaptoethanol, and 5% DMSO. The exact protein concentrations were determined by HPLC amino acid analysis. Typically, 9 μL aliquots of 130 μM ligand were injected stepwise into a sample cell containing 1.43 mL of 10 μM Gal-3C protein until saturation was achieved. For weaker binders, the following titration conditions were used: 780 μM thiodigalactoside with 57 μM Gal-3C protein, 1.2 mM thiolactose with 106 μM Gal-3C protein, and 390 μM ligand **1** with 27 μM Gal-3C protein. Each experimental titration was accompanied by a corresponding control experiment, where solely the ligand was injected into the buffer. Thermodynamic parameters were determined using MicroCal software implemented in Origin 7.0 (MicroCal Inc./Malvern Instruments Ltd., UK). All binding parameters were extracted by fitting the experimental data to a one-site binding model. To estimate whether binding of ligand to the protein was accompanied by proton transfer, titrations in two buffers with different ionization enthalpies (Tris−HCl and HEPES) were performed.[64,65]

NMR Spectroscopy

Ligand-Observed Experiments. ^{19}F NMR spectra of compounds **1**−**8**, both in the presence and absence of Gal-3C, were recorded using a Bruker AVANCE III HD 500 spectrometer (^1H at 500.0 MHz, ^{19}F at 470.4 MHz) at 25 °C. Hexafluorobenzene (δ = −164.9 ppm) was used as an external standard for ^{19}F chemical shift referencing. Stock solutions for ^{19}F NMR titrations were prepared by dissolving compounds **1**, **2**, **3**, **4**, **5**, **6**, and **8** in D_2O, and compound **7** (obtained from Biosynth) in DMSO-d_6. NMR titrations were performed in 5 mm NMR tubes by gradually adding aliquots of the stock solutions to a 200 μM solution of Gal-3C in PBS buffer (pH 7.5). Spectra were collected at different protein saturation levels (protein-to-ligand ratios from 1:0.25 to 1:1.5) to ensure the appearance of distinct signals for the excess unbound ligand. To distinguish the chemical shift of the free ligand during titration, control solutions were prepared containing only the ligand in PBS, supplemented with either D_2O or DMSO-d_6 to match the titration conditions. In addition to the expected signals corresponding to the bound and free forms of the ligands, some spectra exhibited signals at approximately −119.5 ppm and −75.6 ppm. Control experiments identified these as impurities leaching from the glass NMR tubes.[66] It was determined that these impurities, found at micromolar concentrations, did not affect the outcomes of our experiments.

Protein-Observed Experiments. Protein-observed NMR experiments were recorded at 25 °C on a Bruker Avance III HD 850 MHz spectrometer equipped with a triple-resonance ($^{15}N/^{13}C/^1H$) cryoprobe. The protein backbone and partial arginine side-chain assignments of the ^1H,^{15}N, and ^{13}C resonances were obtained using a sample containing 625 μM uniformly ^{15}N,^{13}C-labeled Gal-3C in PBS buffer with 10% D_2O (pH 6.5) at a total volume of 350 μL in a 5 mm Shigemi tube. Standard Bruker 2D and 3D triple-resonance experiments: HN, HNCO, HN(CA)CO, HNCACB, CBCA(CO)-NH, HBHA(CO)NH, (H)CC(CO)NH, HCCH-TOCSY, *N*-edited and C-edited NOESY were acquired. The total experimental time was 12 days. Chemical shift changes upon interaction of Gal-3C with

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ligands **3**, **4** and **6** were monitored using $^{1}H-^{15}N$ SOFAST-HMQC experiments.[67] Samples containing 160 μL of 39 μM Gal-3C in PBS buffer were prepared with increasing stoichiometric equivalents of ligand dissolved in D_2O (1:0.25, 1:0.5, 1:0.75, 1:1, 1:1.25) and 16 μL of D_2O added to each. All protein-observed NMR spectra were processed in Bruker TopSpin and analyzed in NMRFAM-SPARKY.[68] CSP values derived from 2D $^{1}H-^{15}N$ HSQC experiments were calculated using the following formula: $\mathrm{CSP} = \sqrt{\left[\Delta\delta_H^2 + \left(\frac{1}{B}\Delta\delta_N\right)^2\right]}$ where B was set to 6.4 for glycines and 5.0 for all other amino acids.[52]

Molecular Modeling

MD and REST Simulations. The PDB 5H9P structure was used as a model of the receptor. The protein was prepared using the Protein Preparation Workflow[69] in the Schrödinger Suite (version 2025-1) and then modified with PDBFixer (version 1.11). Initial ligand geometries were built based on the ligand conformation in 5H9P. All simulations were performed in OpenMM (version 8.2.0).[70] The protein was parametrized with the ff14SB[71] force field, and small molecules were parametrized with GAFF-2.11.[72] For each simulation, the system was placed into a cubic box with a buffer of 10 Å and TIP3P waters were added. Long-range interactions were treated using the particle-mesh Ewald (PME) method with a cutoff of 10 Å. All bonds involving hydrogens were constrained. The system was first equilibrated for 0.5 ns under NVT conditions (Langevin thermostat at 300 K), followed by another 0.5 ns under NPT conditions (Monte Carlo barostat at 1 bar). The receptor−ligand complexes were simulated using replica exchange with solute tempering (REST2)[73] where the ligand molecule was treated as the tempered region. The number of replicas (N_{replicas}) was determined by the formula: $N_{\mathrm{replicas}} = 2 + \sqrt{\frac{N_{\mathrm{atoms}}}{3}}$, where N_{atoms} is the number of atoms in the tempered region (number of ligand atoms). The temperatures for the individual replicas were spaced exponentially between 300 and 500 K. Each replica was simulated with a 4 fs time step, with swaps between replicas attempted every 1 ps. The baseline replica at 300 K was then extracted and used as the holo trajectory. Three independent REST simulations were performed, each for 200 ns. The apo receptor was simulated using plain molecular dynamics (MD) with a time step of 3 fs and three simulations of length 1000 ns. The three REST2 (or MD) simulations of each system were combined and pruned to 1000- and 50-frame ensembles for downstream processing as described below. The pose retention during the simulations was checked visually by inspecting the produced ensembles, and by calculating ligand heavy-atom RMSD (see Supporting Information).

Interaction Energy Calculation and Decomposition

Interactions between the protein and ligand were quantified in OpenMM (version 8.2.0) by calculating the gas-phase interaction energy ΔE_{int}.[70] The interaction energy was derived from the nonbonded forces in the force field and consists of Coulomb electrostatic interactions and the Lennard-Jones interactions. Geometries for calculating ΔE_{int} were extracted from the holo trajectories in explicit solvent and used without further minimization. Interaction energies were decomposed into per-atom contributions and used as a proxy to quantify the noncovalent contacts between ligand and different regions of the binding pocket. The interaction energy between protein atom i and the ligand was calculated as

$$\Delta E_{\mathrm{int}} = \langle E_{i+\mathrm{ligand}}\rangle - (\langle E_i\rangle + \langle E_{\mathrm{ligand}}\rangle)$$

where $E_{i+\mathrm{ligand}}$ is the force field energy of protein atom i and the ligand, E_i is the energy of protein atom i alone, and E_{ligand} is the energy of the ligand alone. Angle brackets indicate averaging over 50 equally spaced simulation frames.

Calculation of Holoprotein Conformational Changes

Conformational changes of the protein induced by ligand binding were quantified by calculating the perturbation of the position of each protein atom upon ligand binding. All trajectory processing was performed in MDTraj (version 1.9.8).[74] First, all simulation frames were superimposed on the initial protein structure by aligning the C_α carbon atoms. The conformational perturbation of each atom i was expressed as the Euclidean distance (d_i) between its time-averaged positions in the holo and apo trajectories

$$d_i = \|\langle \mathbf{r}_i^{\mathrm{holo}}\rangle - \langle \mathbf{r}_i^{\mathrm{apo}}\rangle\|$$

where $\mathbf{r}_i = (x_i, y_i, z_i)$ denotes the position vector of atom i, and angle brackets indicate averaging over simulation frames. The averages were obtained from 1000 frames in both holo and apo trajectories.

Alchemical FEP Calculations

Affinities of the individual binding poses were calculated as relative binding free energies (RBFE) using alchemical free energy perturbation (FEP). This quantity is denoted here as $\Delta G_{\mathrm{bind}}^{\mathrm{FEP}}$. All FEP calculations were performed in OpenFE (version 1.9.1) using the defaults hybrid topology RBFE protocol.[59] All edges were simulated using 22 λ windows and 25 ns of sampling per window. Each edge was calculated in 3 independent repeats. Protein was parametrized with the Amber ff14SB[71] force field, and small molecules were parametrized with openff-2.2.1[75,76] force field and AM1-BCC[77,78] partial charges. To connect the ligands in both thiodigalactoside series (symmetric binding) and thiolactose series (asymmetric binding) by RBFE edges, a ligand network was constructed for each of the two series. The edges in both networks were enumerated by a custom edge list to optimally connect similar ligands in both networks. For transformations between ligands with large structural differences, we have used intermediate "filler" molecules, which enabled us to span these difficult transformations. The resulting relative binding free energies were aggregated using maximum likelihood estimate (MLE) and referenced to experimental affinities using the ITC data. See the Supporting Information for the details of the network construction, and raw simulated free energy values.

For compounds **2**, **4**, **5**, **6**, and **8**, which are assumed to bind in only a single pose, the binding free energy of this pose was considered the overall binding free energy for the given compound. Compound **7** was assumed to bind in two degenerate orientations (corresponding to the 7-BCDE pose), which are physically indistinguishable and therefore represent an identical thermodynamic state. Accordingly, only one binding free energy was considered for compound **7**. For compounds **1** and **3**, where two nonequivalent binding poses occur, the overall binding free energy $\Delta G_{\mathrm{bind,\ overall}}^{\mathrm{FEP}}$ was determined[9,10] as

$$\Delta G_{\mathrm{bind,overall}}^{\mathrm{FEP}} = -RT\ln\left[\exp\left(-\frac{\Delta G_{\mathrm{bind},i}^{\mathrm{FEP}}}{RT}\right) + \exp\left(-\frac{\Delta G_{\mathrm{bind},j}^{\mathrm{FEP}}}{RT}\right)\right]$$

where R is the gas constant and T is the absolute temperature. The symbols $\Delta G_{\mathrm{bind},i}^{\mathrm{FEP}}$ and $\Delta G_{\mathrm{bind},j}^{\mathrm{FEP}}$ represent the calculated binding free energies of the individual binding poses i and j (**1**-BC and **1**-DE for compound **1**; **3**-BCD and **3**-CDE for compound **3**).

If index i denotes the more favorable binding pose of the two ($\Delta G_{\mathrm{bind},i}^{\mathrm{FEP}} < \Delta G_{\mathrm{bind},j}^{\mathrm{FEP}}$), then the configurational free energy contribution $\Delta\Delta G_{\mathrm{config}}$ arising from the two different binding configurations can be expressed as

$$\Delta\Delta G_{\mathrm{config}} = \Delta G_{\mathrm{bind,overall}}^{\mathrm{FEP}} - \Delta G_{\mathrm{bind},i}^{\mathrm{FEP}} = -RT\ln\left[1 + \exp\left(-\frac{\Delta G_{\mathrm{bind},j}^{\mathrm{FEP}} - \Delta G_{\mathrm{bind},i}^{\mathrm{FEP}}}{RT}\right)\right]$$

■ ASSOCIATED CONTENT

Data Availability Statement

The assignments of the ^{1}H, ^{13}C, and ^{15}N resonances of the carbohydrate recognition domain of human galectin-3 used in this study have been deposited in the Biological Magnetic Resonance Data Bank (BMRB) under the entry 53387. Raw NMR data from ^{19}F NMR titrations and the SPARKY project, consisting of all NMR spectra used for assignment of Gal-3C

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backbone signals, are available on OSF (10.17605/OSF.IO/RW9QE). Python scripts for running the MD and REST simulations in OpenMM and analyzing the trajectories, the initial structures used for these simulations, the simulated trajectories, interaction energies, and holo-apo distances are available on Zenodo (10.5281/zenodo.17559338).

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jmedchem.6c00031.

Binding curves from competitive fluorescence polarization, ITC thermograms, detailed data from free energy calculations, full spectra from ligand-observed ^{19}F NMR titrations, full spectra from protein-observed ^{1}H−^{15}N HSQC NMR experiments, synthetic procedures and analytical data for all synthesized compounds (PDF)

PDB structures of the receptor and ligand poses (ZIP)

Molecular formula strings (CSV)

AUTHOR INFORMATION

Corresponding Authors

Nina Habanová − *Institute of Organic Chemistry and Biochemistry, Czech Academy of Sciences, 160 00 Prague, Czech Republic; Department of Analytical Chemistry, University of Chemistry and Technology Prague, 166 28 Prague, Czech Republic;* orcid.org/0000-0001-6412-9901; Email: nina.habanova@uochb.cas.cz

Jan Choutka − *Institute of Organic Chemistry and Biochemistry, Czech Academy of Sciences, 160 00 Prague, Czech Republic;* orcid.org/0000-0001-6290-9787; Email: jan.choutka@uochb.cas.cz

Kamil Parkan − *Department of Chemistry of Natural Compounds, University of Chemistry and Technology Prague, 166 28 Prague, Czech Republic;* orcid.org/0000-0001-7585-6004; Email: Kamil.Parkan@vscht.cz

Authors

Jakub Zýka − *Department of Chemistry of Natural Compounds, University of Chemistry and Technology Prague, 166 28 Prague, Czech Republic*

Vít Prouza − *Department of Chemistry of Natural Compounds, University of Chemistry and Technology Prague, 166 28 Prague, Czech Republic*

Jaroslav Kozák − *Institute of Organic Chemistry and Biochemistry, Czech Academy of Sciences, 160 00 Prague, Czech Republic*

Martin Zavřel − *Institute of Organic Chemistry and Biochemistry, Czech Academy of Sciences, 160 00 Prague, Czech Republic*

Milan Kožíšek − *Institute of Organic Chemistry and Biochemistry, Czech Academy of Sciences, 160 00 Prague, Czech Republic;* orcid.org/0000-0002-9476-3409

Pavel Srb − *Institute of Organic Chemistry and Biochemistry, Czech Academy of Sciences, 160 00 Prague, Czech Republic*

Radek Pohl − *Institute of Organic Chemistry and Biochemistry, Czech Academy of Sciences, 160 00 Prague, Czech Republic;* orcid.org/0000-0001-7898-946X

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.jmedchem.6c00031

Author Contributions

[‖]N.H., J.Z., and J.C. contributed equally to this work. N.H., J.Z., and J.C. designed the study; N.H. and R.P. performed NMR spectroscopy of small compounds; N.H. and P.S. performed protein NMR spectroscopy; J.Z. and V.P. synthesized small compounds; M.Z. produced and purified Gal-3C; M.K. performed the ITC experiments; J.K. performed the FPA experiments; J.C. performed the in silico modeling; J.C. wrote the paper with input from all authors; K.P. and R.P. acquired funding and supervised the study.

Notes

The authors declare no competing financial interest.

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ABBREVIATIONS

ACN, acetonitrile; APT, attached proton test; BMRB, Biological Magnetic Resonance Data Bank; cryo-EM, cryogenic electron microscopy; CSP, chemical shift perturbation; DIPEA, *N*,*N*-diisopropylethylamine; FEP, free energy perturbation; FPA, fluorescence polarization assay; HEPES, 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid; IPTG, isopropyl β-D-1-thiogalactopyranoside; ITC, isothermal titration calorimetry; Lac, lactose; LacNAc, *N*-acetyllactosamine; MBAR, multistate Bennett acceptance ratio; MWCO, molecular weight cutoff; Nf, nonafluorobutanesulfonyl; NfN_3, nonafluorobutanesulfonyl azide; NPT, constant particle number, pressure, and temperature ensemble; NVT, constant particle number, volume, and temperature ensemble; PME, particle-mesh Ewald; PMP, para-methoxyphenyl; REST, replica exchange with solute tempering; SOFAST-HMQC, band-selective optimized flip-angle short-transient heteronuclear multiple quantum coherence; TMSOTf, trimethylsilyl trifluoromethanesulfonate; TOCSY, total correlation spectroscopy.

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